Math **163** Name (PLEASE PRINT) Cap'n Orangebeard

Exam **1C** **2020 Fall**/Spring/Summer I /Summer II Score _____

Please fill in the following table checking boxes corresponding to parts of problems that you have completed and submitted and an expected grade. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, as per the directions of the problem. Ideally you should know exactly what you have done correctly and what grade you should receive. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

| | | | | | | | | | | Expected | Received |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Part | Part | Part | Part | Part | Part | Part | Part | Part | Grade | Grade |
| | (i) | (ii) | (iii) | (iv) | (v) | (vi) | (vii) | (viii) | (ix) | | |
| Number | | | | | | | | | | | |
| | | | | | | | | | | | |
| 1a | 2 | 2 | 2 | | | | | | | 6 | |
| 1b | 3 | 3 | | | | | | | | 6 | |
| 1c | | | | | | | | | | | |
| | | | | | | | | | | | |
| 2a | 3 | 3 | | | | | | | | 6 | |
| 2b | 6 | | | | | | | | | 6 | |
| 2c | | | | | | | | | | | |
| | | | | | | | | | | | |
| 3a | 1 | 1 | 1 | 1 | 1 | 1 | | | | 6 | |
| 3b | 6 | | | | | | | | | 6 | |
| 3c | | | | | | | | | | | |
| | | | | | | | | | | | |
| | | | | | | | | Total | | 36 | |
| | | | | | | | | | | | |

$JN := (J_k \in 0..9 | k \in 1..8) =$ 40130044

Hence

$J_1 = 4 \quad J_2 = 0 \quad J_3 = 1 \quad J_4 = 3 \quad J_5 = 0 \quad J_6 = 0 \quad J_7 = 4 \quad J_8 = 4$

**R** Write **the elements of** every numerical set or multiset appearing in this exam **in increasing order.**

**D** $\mathcal{U} := 0..9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

**D** $SJ := St(J) := \left\{ J_k \in 0..9 \middle| k \in 1..8 \right\} = \{ 0, 1, 3, 4 \}$

$\nu(SJ) = 4$

**D** $Lst(J) := \left( (SJ)_k \in 0..9 \middle| k \in 1..(\nu(SJ)) \right) = (0, 1, 3, 4)$

**D** $\forall l \in 1..(\nu(SJ)) \quad n_l := \nu\left( \left\{ k \in 1..8 \middle| J_k = (SJ)_l \right\} \right)$

**D** $MSJ := MltSt(J) := \left\{ (SJ)_m : n_m \middle| m \in 1..(\nu(SJ)) \right\}$

$= \{ 0:3, 1:1, 3:1, 4:3 \}$

**D** $\nu(MSJ) = \sum_{m \in 1..(\nu(SJ))} n_m = 8$

**D** $s := mnm_{k \in 1..8}\{J_k | J_k \neq 0\} = 1$

**D** $b := mxm_{k \in 1..8}\{J_k | k \in 1..8\} = 4$

**D** Define Y := Yes N := No Pf := Proof W := Witness

**D** $\forall x \in \mathbb{Z}$ $Evn(x)$ $:\Leftrightarrow$ $x$ is even.

**D** $\forall x \in \mathbb{Z}$ $Odd(x)$ $:\Leftrightarrow$ $x$ is odd.

**D** $\forall x \in \mathbb{N}$ $Prm(x)$ $:\Leftrightarrow$ $x$ is prime.

**R** **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in any set in the above. **So, if you do not have the correct sets, all your answers will be wrong.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

**R** **You will lose points for making any extraneous mark on the exam.**

**R** You may only submit complete answers to questions marked explicitly on the exam. You will lose points for unfinished work.

**R** Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper drawn with a ruler. You will lose points for otherwise.

**R** Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

6 (1a) Compute, $\forall n \in \mathbb{N}, \forall k \in 0..n$ the following, and express the result as a polynomial in n. **Show computations or substitutions into appropriate formulas exactly as shown in class on the facing page and write the answers on this page.**

2 (i) $C\left(C\left(v\left(St\left(J\right)\right),0\right),P\left(v\left(MltSt\left(J\right)\right),0\right)\right)$

$=$ |

2 (ii) $C\left(P\left(v\left(MltSt\left(J\right)\right),v\left(St\left(J\right)\right)\right),P\left(v\left(MltSt\left(J\right)\right),v\left(St\left(J\right)\right)\right)\right)$

$=$ /

2 (iii) $C\left(P\left(v\left(MltSt\left(J\right)\right),v\left(MltSt\left(J\right)\right)\right),P\left(v\left(St\left(J\right)\right),v\left(St\left(J\right)\right)\right)\right)$

$= \binom{(8!)}{(4!)}$

1a.i.0

$$\frac{C(n,0)=1}{C(v(St(J)),0)=1}\left\langle n \leftarrow v\left(St\left(J\right)\right)\right\rangle$$

1a.i.1

$$\frac{P(n,0)=1}{P(v(MltSt(J)),0)=1}\left\langle n \leftarrow v\left(MltSt\left(J\right)\right)\right\rangle$$

1a.i.2

$$C\left(C\left(v\left(St\left(J\right)\right),0\right),P\left(v\left(MltSt\left(J\right)\right),0\right)\right)$$

$$= C(1,1)$$

$$= 1$$

1a.ii

$$\frac{C(n,n)=1}{C\left(P\left(v\left(MltSt(J)\right),v\left(St(J)\right)\right),P\left(v\left(MltSt(J)\right),v\left(St(J)\right)\right)\right)=1}\left\langle n \leftarrow P(v(MltSt(J)),v(St(J)))\right\rangle$$

1a.iii.0

$$\frac{P(n,n)=n!}{P\left(v\left(MltSt\left(J\right)\right),v\left(MltSt\left(J\right)\right)\right)=(v(MltSt(J)))!}\left\langle n \leftarrow v(MltSt(J))\right\rangle$$

1a.iii.1

$$\frac{P(n,n)=n!}{P(v(St(J)),v(St(J)))=(v(St(J)))!}\left\langle n \leftarrow v(St(J))\right\rangle$$

1a.iii.2

$$C\left(P\left(v\left(MltSt\left(J\right)\right),v\left(MltSt\left(J\right)\right)\right),P\left(v\left(St\left(J\right)\right),v\left(St\left(J\right)\right)\right)\right)$$

$$= C(((v(MltSt(J)))!),((v(St(J)))!))$$

$$= C((8!),(4!))$$

$$= \binom{(8!)}{(4!)}$$

6 (1b) **Provide complete reasoning by making a systematic, exhaustive list in the form of a table,** to generate, list, and count the following:

3 (i) D $F \coloneqq \left\{ (x,y) \in \left( St\left( J \right) \right) \times \left( St\left( J \right) \right) \middle| x|y \right\}$

compute: $F$ = $\{$ (0,0), (1,0), (1,1), (1,3), (1,4), (3,0), (3,3), (4,0), (4,4) $\}$

compute: $\nu\left( F \right) =$ 9

3 (ii) D $G \coloneqq \left\{ (x,y) \in \left( St\left( J \right) \right) \times \left( St\left( J \right) \right) \middle| y|x \right\}$

compute: $G$ = $\{$ (0,0), (0,1), (0,3), (0,4), (1,1), (3,1), (3,3), (4,1), (4,4) $\}$

compute: $\nu\left( G \right) =$ 9

16. i

$$F := \{(x,y) \in (St(J)) \times (St(J)) \mid x \mid y\}$$

| $(x,y)$ | 0 | 1 | 3 | 4 |
|---|---|---|---|---|
| 0 | (0,0) | (0,1) | (0,3) | (0,4) |
| 1 | (1,0) | (1,1) | (1,3) | (1,4) |
| 3 | (3,0) | (3,1) | (3,3) | (3,4) |
| 4 | (4,0) | (4,1) | (4,3) | (4,4) |

$$F = \{(0,0), (1,0), (1,1), (1,3), (1,4), (3,0), (3,3), (4,0), (4,4)\}$$

16. ii

$$G := \{(x,y) \in (St(J)) \times (St(J)) \mid y \mid x\}$$

| $(x,y)$ | 0 | 1 | 3 | 4 |
|---|---|---|---|---|
| 0 | (0,0) | (0,1) | (0,3) | (0,4) |
| 1 | (1,0) | (1,1) | (1,3) | (1,4) |
| 3 | (3,0) | (3,1) | (3,3) | (3,4) |
| 4 | (4,0) | (4,1) | (4,3) | (4,4) |

$$G = \{(0,0), (0,1), (0,3), (0,4), (1,1), (3,1), (3,3), (4,1), (4,4)\}$$

6 (2a) **Use a tree** (**which you should draw on the facing page**) to **generate**, **list**, and **count** the elements of the set of all **2-letter words** in the alphabet $\{b-s, b, b+s\}$ such that

My $\{b-s, b, b+s\} = \{ 3, 4, 5 \}$

3 (i) order matters and repetition is not allowed, and

$$Wrd\left(2, \{3, 4, 5\}, OM, \cancel{RA}\right) = \{ 34, 35, 43, 45, 53, 54 \}$$

$$v\left(Wrd\left(2, \{3, 4, 5\}, OM, \cancel{RA}\right)\right) = 6$$

3 (ii) order does **not** matter and repetition is not allowed

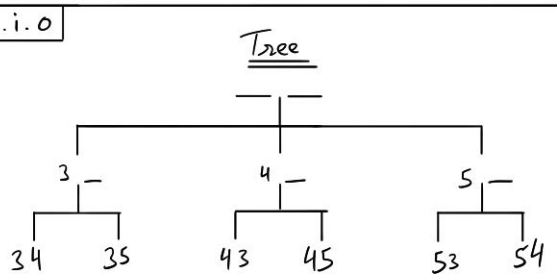
$$Wrd\left(2, \{3, 4, 5\}, \cancel{OM}, \cancel{RA}\right) = \{ 34, 35, 45 \}$$

$$v\left(Wrd\left(2, \{3, 4, 5\}, \cancel{OM}, \cancel{RA}\right)\right) = 3$$

2a

$$\{b-s,\ b,\ b+s\}$$
$$= (\{b-s,\ b,\ b+s\})\left\langle \begin{matrix} b \leftarrow 4 \\ s \leftarrow 1 \end{matrix} \right\rangle$$
$$= \{4-1,\ 4,\ 4+1\}$$
$$= \{3,\ 4,\ 5\}$$

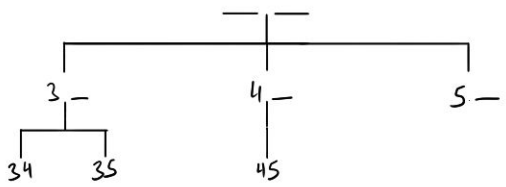
2a.i.0




2a.i.1

$$Wrd(2, \{3,4,5\}, OM, \cancel{RA}) = \begin{Bmatrix} 34, & 35, \\ 43, & 45, \\ 53, & 54 \end{Bmatrix}$$

2a.ii.0




2a.ii.1

$$Wrd(2, \{3,4,5\}, \cancel{OM}, \cancel{RA}) = \begin{Bmatrix} 34, & 35, \\ 45 \end{Bmatrix}$$

6 (2b) Using a suitable tree, **generate, count, and list** the elements of the set $S$ of **distinct 3-digit** numbers that may be made by **using the elements of** $St\left( J \right)$ **as digits**:

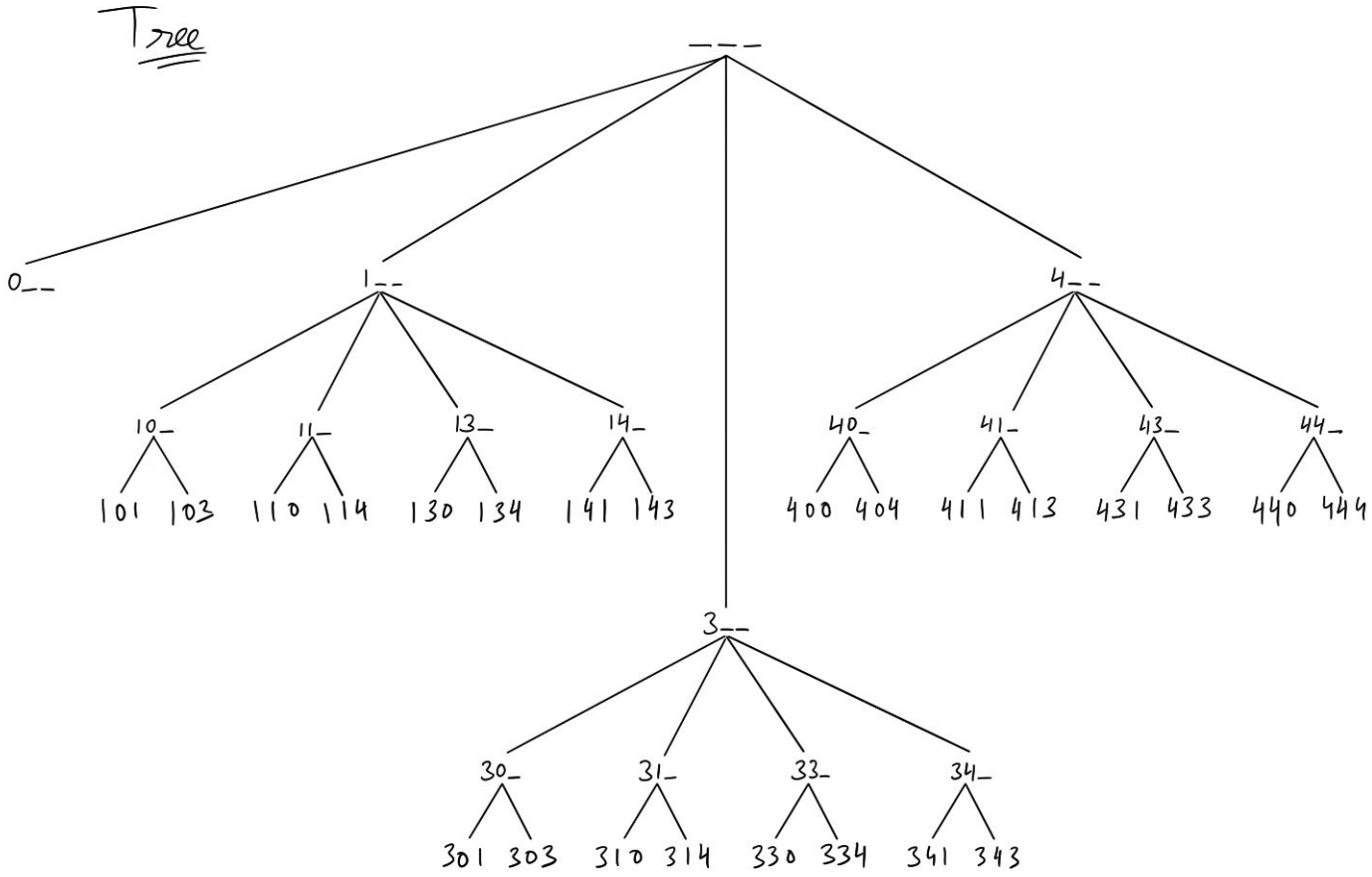
$$S = \left\{ \begin{array}{l} 101, 103, 110, 114, 130, 134, 141, 143, \\ 301, 303, 310, 314, 330, 334, 341, 343, \\ 400, 404, 411, 413, 431, 433, 440, 444 \end{array} \right\}$$

**such that:**

- The digits of the 3-digit number add up to an even number.

**E** The 3-digit number 121 satisfies the condition since $1 + 2 + 1 = 4$ which is an even number while the three-digit number 111 does not satisfy the condition since $1 + 1 + 1 = 3$ which is an odd number.

26.0




26.1

$$S = \begin{Bmatrix} 101, 103, 110, 114, 130, 134, 141, 143, \\ 301, 303, 310, 314, 330, 334, 341, 343, \\ 400, 404, 411, 413, 431, 433, 440, 444 \end{Bmatrix}$$

6 (3a) **Fill in the blanks appropriately** and answer the following:

$s = 1$ $b = 4$ $b + s = 5$ $b - s = 3$

1 (i) Compute, using the **binomial theorem**:

$$BnmlCfn\left(x^{b-s}y^{b+s}, (x+y)^{(2b)}\right) = BnmlCfn\left(x^{(3)}y^{(5)}, (x+y)^{(8)}\right) = \binom{8}{5}$$

1 (ii) Compute, using the **binomial theorem**:

$$BnmlCfn\left(x^{b+s}y^{b-s}, (x+y)^{(2b)}\right) = BnmlCfn\left(x^{(5)}y^{(3)}, (x+y)^{(8)}\right) = \binom{8}{3}$$

1 (iii) Compute, using the **multinomial theorem**:

$$MltnmlCfn\left(x^{b-s}y^{b+s}, (x+y)^{(2b)}\right) = MltnmlCfn\left(x^{(3)}y^{(5)}, (x+y)^{(8)}\right) = \binom{8}{3\ 5}$$

1 (iv) Compute, using the **multinomial theorem**:

$$MltnmlCfn\left(x^{b+s}y^{b-s}, (x+y)^{(2b)}\right) = MltnmlCfn\left(x^{(5)}y^{(3)}, (x+y)^{(8)}\right) = \binom{8}{5\ 3}$$

1 (v) The answers in (i), (ii), (iii) and (iv) are equal; record answer on this page and work on the facing page:

(Y) N (Pf) W

1 (vi) The answer to (v) depend on the values of $s$ and of $b$; record answer on this page and work on the facing page:

Y (N) (Pf) W

3a

$$\frac{\frac{\frac{b=4 \quad n-k=b-s \quad k=b+s}{n-(b+s)=b-s}}{n=(b-s)+(b+s)}}{\begin{matrix} n \\ =(2)(b) \\ =(2)(4) \\ =8 \end{matrix}}$$

3a.i

$$\begin{aligned} & \mathrm{BnmlCfn}\left(x^3 y^5,\ (x+y)^8\right) \\ = \ & \mathrm{BnmlCfn}\left(x^{8-5} y^5,\ (x+y)^8\right) \\ = \ & \left(\mathrm{BnmlCfut}\left(a^{n-k} b^k,\ (a+b)^n\right)\right)\left\langle \begin{matrix} n \leftarrow 8 \\ k \leftarrow 5 \\ a \leftarrow x \\ b \leftarrow y \end{matrix} \right\rangle \\ = \ & \left(\left(\begin{matrix} n \\ k \end{matrix}\right)\right)\langle '' \rangle \\ = \ & \begin{pmatrix} 8 \\ 5 \end{pmatrix} \end{aligned}$$

3a.ii

$$\begin{aligned} & \mathrm{BnmlCfn}\left(x^5 y^3,\ (x+y)^8\right) \\ = \ & \mathrm{BnmlCfn}\left(x^{8-3} y^3,\ (x+y)^8\right) \\ = \ & \left(\mathrm{BnmlCfut}\left(a^{n-k} b^k,\ (a+b)^n\right)\right)\left\langle \begin{matrix} n \leftarrow 8 \\ k \leftarrow 3 \\ a \leftarrow x \\ b \leftarrow y \end{matrix} \right\rangle \\ = \ & \left(\left(\begin{matrix} n \\ k \end{matrix}\right)\right)\langle '' \rangle \\ = \ & \begin{pmatrix} 8 \\ 3 \end{pmatrix} \end{aligned}$$

3a.iii

$$\begin{aligned} & \mathrm{MltnmlCfn}\left(x^3 y^5,\ (x+y)^8\right) \\ = \ & \left(\mathrm{MltnmlCfut}\left(\left(x_1^{n_1}\right)\left(x_2^{n_2}\right)\ldots\left(x_p^{n_p}\right),\ \left(x_1+x_2+\cdots+x_p\right)^n\right)\right)\left\langle \begin{matrix} p \leftarrow 2 \\ n \leftarrow 8 \\ n_1 \leftarrow 3 \\ n_2 \leftarrow 5 \\ x_1 \leftarrow x \\ x_2 \leftarrow y \end{matrix} \right\rangle \\ = \ & \left(\left(\begin{matrix} n \\ n_1 \ n_2 \cdots n_p \end{matrix}\right)\right)\langle '' \rangle \\ = \ & \begin{pmatrix} 8 \\ 3 \ 5 \end{pmatrix} \end{aligned}$$

3a.iv

$$\begin{aligned} & \mathrm{MltnmlCfn}\left(x^5 y^3,\ (x+y)^8\right) \\ = \ & \left(\mathrm{MltnmlCfut}\left(\left(x_1^{n_1}\right)\left(x_2^{n_2}\right)\ldots\left(x_p^{n_p}\right),\ \left(x_1+x_2+\cdots+x_p\right)^n\right)\right)\left\langle \begin{matrix} p \leftarrow 2 \\ n \leftarrow 8 \\ n_1 \leftarrow 5 \\ n_2 \leftarrow 3 \\ x_1 \leftarrow x \\ x_2 \leftarrow y \end{matrix} \right\rangle \\ = \ & \left(\left(\begin{matrix} n \\ n_1 \ n_2 \cdots n_p \end{matrix}\right)\right)\langle '' \rangle \\ = \ & \begin{pmatrix} 8 \\ 5 \ 3 \end{pmatrix} \end{aligned}$$

3a.v.0

$$\begin{aligned} & \binom{8}{5} \\ = \ & \frac{8!}{(5!)((8-5)!)} \\ = \ & \frac{8!}{(5!)(3!)} \end{aligned}$$

3a.v.1

$$\begin{aligned} & \binom{8}{3} \\ = \ & \frac{8!}{(3!)((8-3)!)} \\ = \ & \frac{8!}{(3!)(5!)} \end{aligned}$$

3a.v.2

$$\begin{aligned} & \begin{pmatrix} 8 \\ 3 \ 5 \end{pmatrix} \\ = \ & \begin{pmatrix} 8 \\ 5 \ 3 \end{pmatrix} \\ = \ & \frac{8!}{(3!)(5!)} \end{aligned}$$

3a.v.3

$$\begin{aligned} & \binom{8}{5} \\ = \ & \binom{8}{3} \\ = \ & \begin{pmatrix} 8 \\ 3 \ 5 \end{pmatrix} \\ = \ & \begin{pmatrix} 8 \\ 5 \ 3 \end{pmatrix} \end{aligned}$$

3a.vi.0

$$\begin{aligned} & \binom{(2)(b)}{b+s} \\ = \ & \frac{((2)(b))!}{((b+s)!)((2b-(b+s))!)} \\ = \ & \frac{((2)(b))!}{((b+s)!)((b-s)!)} \end{aligned}$$

3a.vi.1

$$\begin{aligned} & \binom{(2)(b)}{b-s} \\ = \ & \frac{((2)(b))!}{((b-s)!)((2b-(b-s))!)} \\ = \ & \frac{((2)(b))!}{((b-s)!)((b+s)!)} \end{aligned}$$

3a.vi.2

$$\begin{aligned} & \begin{pmatrix} (2)(b) \\ b-s \ \ b+s \end{pmatrix} \\ = \ & \begin{pmatrix} (2)(b) \\ b+s \ \ b-s \end{pmatrix} \\ = \ & \frac{((2)(b))!}{((b-s)!)((b+s)!)} \end{aligned}$$

3a.vi.3

$$\begin{aligned} & \binom{(2)(b)}{b+s} \\ = \ & \binom{(2)(b)}{b-s} \\ = \ & \begin{pmatrix} (2)(b) \\ b-s \ \ b+s \end{pmatrix} \\ = \ & \begin{pmatrix} (2)(b) \\ b+s \ \ b-s \end{pmatrix} \end{aligned}$$

6 (3b) Fill in the blank appropriately, and show your work on the facing side to compute the value of:

$$\sum_{p+q=2} \binom{4}{p \quad 2 \quad q}$$

$= \quad (2)(3!) + \left(\frac{4!}{2}\right)$

3b.0

$$\frac{n = n_1 + n_2 + n_3}{n = p + 2 + q} \left\langle \begin{matrix} n_1 \leftarrow p \\ n_2 \leftarrow 2 \\ n_3 \leftarrow q \end{matrix} \right\rangle$$

$$\frac{n = 2 + (p+q)}{n = 2 + 2} \left\langle (p+q) \leftarrow 2 \right\rangle$$

$$n = 4$$

3b.1

$$4 = 0 + 2 + 2 = 1 + 2 + 1$$
$$= 2 + 2 + 0$$

3b.2

$$\sum_{p+q=2} \binom{4}{p\ 2\ q}$$

$$= \begin{vmatrix} \binom{4}{0\ 2\ 2} \\ + \binom{4}{1\ 2\ 1} \\ + \binom{4}{2\ 2\ 0} \end{vmatrix} = \begin{vmatrix} 3! \\ + \frac{4!}{2} \\ + 3! \end{vmatrix} = (2)(3!) + \left(\frac{4!}{2}\right)$$

3b.3

$$\binom{4}{0\ 2\ 2}$$

$$= \binom{4}{2\ 2\ 0}$$

$$= \frac{4!}{(0!)(2!)(2!)}$$

$$= \frac{\cancel{(4)}(3)(2)}{(1)\cancel{(2)}\cancel{(2)}}$$

$$= 3!$$

3b.4

$$\binom{4}{1\ 2\ 1}$$

$$= \frac{4!}{(1!)(2!)(1!)}$$

$$= \frac{(4)(3)\cancel{(2)}}{(1)\cancel{(2)}(1)}$$

$$= \frac{4!}{2}$$